

The Hitchhiking Effect on the Site Frequency Spectrum of DNA Polymorphisms

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ABSTRACT

The level of DNA sequence variation is reduced in regions of the *Drosophila melanogaster* genome where the rate of crossing over per physical distance is also reduced. This observation has been interpreted as support for the simple model of genetic hitchhiking, in which directional selection on rare variants, *e.g.*, newly arising advantageous mutants, sweeps linked neutral alleles to fixation, thus eliminating polymorphisms near the selected site. However, the frequency spectra of segregating sites of several loci from some populations exhibiting reduced levels of nucleotide diversity and reduced numbers of segregating sites did not appear different from what would be expected under a neutral equilibrium model. Specifically, a skew toward an excess of rare sites was not observed in these samples, as measured by Tajima's *D*. Because this skew was predicted by a simple hitchhiking model, yet it had never been expressed quantitatively and compared directly to DNA polymorphism data, this paper investigates the hitchhiking effect on the site frequency spectrum, as measured by Tajima's *D* and several other statistics, using a computer simulation model based on the coalescent process and recurrent hitchhiking events. The results presented here demonstrate that under the simple hitchhiking model (1) the expected value of Tajima's *D* is large and negative (indicating a skew toward rare variants), (2) that Tajima's test has reasonable power to detect a skew in the frequency spectrum for parameters comparable to those from actual data sets, and (3) that the Tajima's *D*s observed in several data sets are very unlikely to have been the result of simple hitchhiking. Consequently, the simple hitchhiking model is not a sufficient explanation for the DNA polymorphism at those loci exhibiting a decreased number of segregating sites yet not exhibiting a skew in the frequency spectrum.

A MAJOR goal of *Drosophila* population genetics during the past ten years has been to determine the distribution of DNA sequence polymorphism in the genome (LANGLEY 1990; KREITMAN 1991; AQUADRO 1992). One important observation made by surveys of natural populations of *D. melanogaster* is that the level of DNA sequence variation, in terms of nucleotide diversity and the number of segregating polymorphic sites (KREITMAN 1991), is reduced in regions of the genome where the rate of crossing over per physical distance is lower than normal (AGUADÉ *et al.* 1989, 1994; MIYASHITA 1990; BEGUN and AQUADRO 1991; BERRY *et al.* 1991; MARTÍN-CAMPOS *et al.* 1992; LANGLEY *et al.* 1993). This observation has been made in two other species, *D. simulans* (BEGIN and AQUADRO 1991; BERRY *et al.* 1991; MARTÍN-CAMPOS *et al.* 1992; LANGLEY *et al.* 1993) and *D. ananassae* (STEPHAN 1989; STEPHAN and LANGLEY 1989; STEPHAN and MITCHELL 1992).

Initially, two hypotheses were proposed to explain this reduction in nucleotide diversity and in the number of segregating sites. First, a reduced total mutation rate

and/or greater functional constraint (*i.e.*, stronger purifying selection) in regions with reduced recombination should decrease the neutral equilibrium heterozygosity (AGUADÉ *et al.* 1989). This neutralist explanation predicts a reduction in interspecific divergence in these regions (KIMURA 1983). When the levels of interspecific divergence at these loci were determined not to be unusually low, this first hypothesis was ruled out (BEGIN and AQUADRO 1991; BERRY *et al.* 1991; MARTÍN-CAMPOS *et al.* 1992; LANGLEY *et al.* 1993). The second proposed explanation of this reduction was the hitchhiking effect hypothesis (AGUADÉ *et al.* 1989; KAPLAN *et al.* 1989; STEPHAN 1989; STEPHAN and LANGLEY 1989). Simple genetic hitchhiking is the process whereby directional selection on rare variants, *e.g.*, newly arising advantageous mutants, sweeps linked neutral alleles to fixation, thus eliminating polymorphisms near the selected site. Analyses of the simple hitchhiking model predict a reduction in heterozygosity (MAYNARD SMITH and HAIGH 1974; STEPHAN *et al.* 1992) and in the number of segregating sites (KAPLAN *et al.* 1989) adjacent to a selected substitution. These effects of a hitchhiking event should cover a greater physical distance and thus be detected easier in regions of restricted crossing over (KAPLAN *et*

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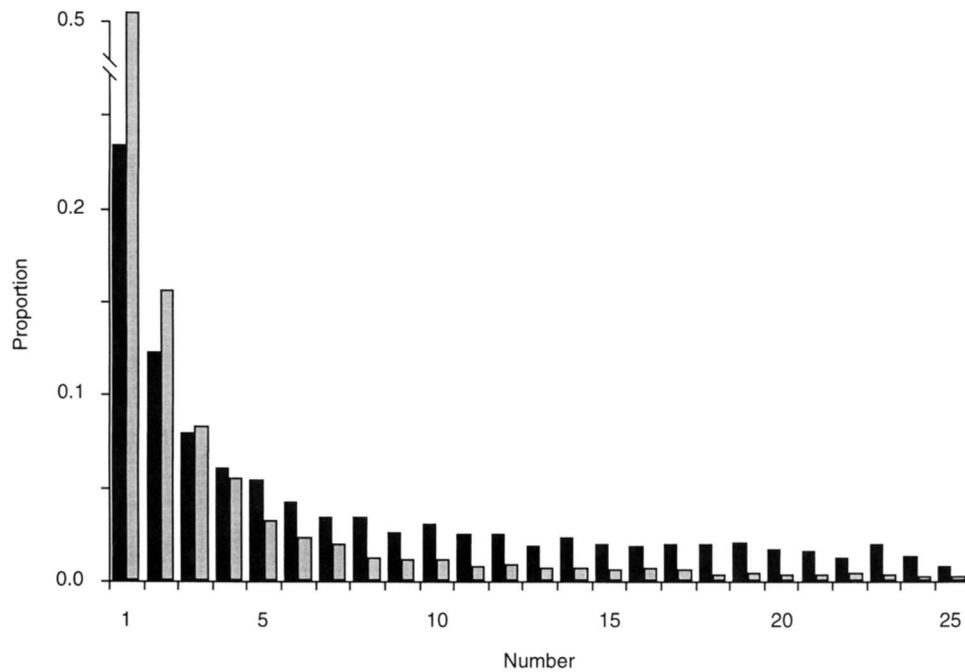


FIGURE 1.—The site frequency spectrum under no hitchhiking (black bars) and under strong hitchhiking (gray bars). The horizontal axis is the number of times the rarer of the two mutational states at a segregating site occurred in a sample. The vertical axis is the proportion of sites that occurred in samples a particular number of times. For example, the proportion of sites among all the simulations with hitchhiking that occurred only one time (uniques) is ~ 0.5 . The parameters used in 1000 simulations were $n = 50$ and $S = 17$ for both neutral and with hitchhiking cases. The hitchhiking cases had $\alpha = 10^5$ and $\Lambda_r = 6.400 \times 10^{-5}$, values chosen because they produced the $R_{E(T)} = 0.32$ observed in the *su(w^o)* data set discussed in Table 2. See text for details.

al. 1989). A corollary to this prediction is the positive correlation between the coefficient of exchange per physical length and the level of nucleotide diversity for many different loci asserted by some authors (BEGUN and AQUADRO 1992; WIEHE and STEPHAN 1993; but see BEGUN *et al.* 1994 for a notable exception).

Another observation made by surveys of regions of the genome with reduced crossing over per physical distance is that the frequency spectrum of segregating sites is often not significantly different from what would be expected under a neutral equilibrium model (MARTÍN-CAMPOS *et al.* 1992; BEGUN and AQUADRO 1993, 1995; AGUADÉ *et al.* 1994). A frequency spectrum is the distribution of the frequencies of segregating sites observed in a sample of alleles or sequences. An example of a simulated neutral frequency spectrum is shown in Figure 1 (black bars). A statistic used to quantitatively describe the frequency spectrum is Tajima's D (TAJIMA 1989). This statistic is based on the normalized difference between two estimators of the neutral parameter, $4Nu$, where N is the number of individuals in a diploid population and u is the neutral mutation rate per generation. Specifically, $\hat{\theta}$, which is based on the number of segregating sites in a sample of sequences (WATTERSON 1975), is subtracted from $\hat{k}$, which is the average number of pairwise differences between sequences (TAJIMA 1983). This difference is normalized so its expectation and variance under neutrality are approximately zero

and one, respectively. Because $\hat{k}$ decreases as the frequencies of a given number of sites in a sample of sequences decrease, while $\hat{\theta}$ does not change, a large negative value of D indicates an excess of rare sites over what would be expected under neutrality.

That some Tajima's D s were not significantly different from zero was surprising because this contradicted the suggestion that the frequency spectrum should be skewed as another effect of hitchhiking. Some authors argued that if a population is constantly recovering from the removal of polymorphism by hitchhiking events, then polymorphisms found in small samples will be rare or unique, since their frequencies increase from zero by the relatively weak forces of mutation and drift (AGUADÉ *et al.* 1989; HUDSON 1990; LANGLEY 1990). Yet no quantitative investigation of the frequency spectrum under hitchhiking (and specifically Tajima's D) was conducted until the present paper (except see brief discussions by LANGLEY 1990; HUDSON 1993). Thus it remained unresolved whether the observations of Tajima's D s not significantly different from zero (*i.e.*, the failure of Tajima's test to reject neutrality) contradicted the predictions of the hitchhiking model. Specifically, these observations could be consistent with the simple hitchhiking model if the distribution of Tajima's D is not strongly skewed in the negative direction under the simple hitchhiking model. If this is the case, then Tajima's test might not have enough power to detect a

skew in the frequency spectrum given the amount of data in particular surveys.

This paper uses a simulation of the simple hitchhiking model to assess these interpretations of nonsignificant Tajima's D s in regions of the *D. melanogaster* genome where the rate of crossing over per physical distance and the number of segregating sites are reduced. We conclude that under the simple hitchhiking model (1) the expected value of Tajima's D is large and negative under hitchhiking, (2) the power of Tajima's test to detect a skew in the frequency spectrum for the data sets available is reasonably large, and (3) the observed values of Tajima's D for actual data sets are very unlikely. The implication of these conclusions is that the simple hitchhiking model is not a sufficient explanation for the observed quantity and distribution of DNA polymorphism in regions of restricted crossing over.

MATERIALS AND METHODS

The simulations described here executed the simple model of genetic hitchhiking with one neutral and one selected locus (MAYNARD SMITH and HAIGH 1974). Genealogical histories of samples of n alleles at the neutral locus were assembled by a computer program written in C language that implemented the standard principles of the neutral coalescent process (HUDSON 1990) in combination with recurrent selected substitutions at a selected locus (KAPLAN *et al.* 1989). These neutral alleles represent a random sample of DNA sequences from a natural population. Figure 2 shows a schematic of one hypothetical realization of a typical genealogy constructed by this simulation. The topology of this genealogy is represented in the computer as an array of data structures, each element of which contains information about a node in the genealogy including links to adjacent nodes. As is standard for coalescent models, the simulation starts at the present (from the bottom of Figure 2) and proceeds backward in time (to the top of Figure 2). The simulation alternates between two phases to construct the genealogies. The first phase is a neutral phase (exemplified on the left side of Figure 2). The code for this portion of the program was written according to HUDSON (1983). During a neutral phase, three possible events can occur after an exponential waiting time: (1) a pair of neutral alleles can coalesce (*i.e.*, they merge at a common ancestor), (2) a neutral allele can recombine intragenically anywhere within its length, or (3) the system can enter a selective phase. Because this model is applied to regions of the genome where the rate of crossing over per physical distance is severely restricted, we only report results for zero intragenic recombination during the neutral phase. Nonzero levels of intragenic recombination do not significantly alter the results presented in this paper anyway (data not shown). This intragenic recombination should not be confused with recombination between the neutral and selected loci, which does occur in these simulations.

A selective phase consists of the substitution of a selectively favored allele linked to the neutral locus. Selective phases are represented by boxes on the right side of Figure 2. Their short time scale relative to that of the neutral tree is indicated by the compression of these boxes into thin lines on the left side of Figure 2. For each substitution, the recombination distance between the two loci (neutral and selected) is drawn randomly from a uniform distribution between zero and M

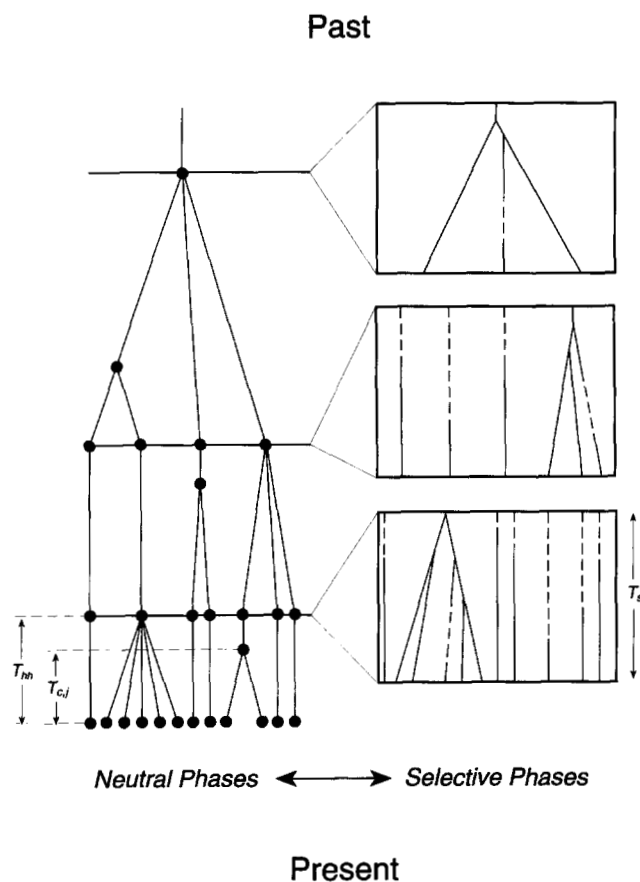


FIGURE 2.—A schematic diagram of one hypothetical realization of a typical hitchhiking genealogy. Depicted here are two different time scales. The neutral phase time scale consists of coalescent events (T_{cj}) and hitchhiking events (T_{hh}). Specifically, for j alleles, $E(T_{cj}) = 1/(j/2)$, and $E(T_{hh}) = (1/\Lambda_r)$. During the selective phases, — represent alleles linked to the selectively favored allele, and --- represent alleles linked to the selectively unfavored allele.

(see Table 1 for a list of symbols). M is the largest recombination distance at which a single hitchhiking event has a detectable effect on $E(T)$. $E(T)$ is the expected total size of a genealogy of a sample of alleles measured in $2N$ generations, calculated from the simulations by averaging the genealogy sizes from many iterations. M was determined empirically to equal α .

Because the simulation works backward in time, the selectively favored allele is fixed at the start of a selective phase, thus all the extant neutral alleles at the beginning are linked to the favored allele. Denoting b as the wild-type allele and B as the favored allele at the selected locus, the three genotypes, bb , bB , and BB , have fitnesses 1, $1 + s$, and $1 + 2s$, respectively (MAYNARD SMITH and HAIGH 1974; KAPLAN *et al.* 1989; STEPHAN *et al.* 1992), where s is the selective advantage of the favored allele. Under the action of additive directional selection of strength $\alpha = 2Ns$, where s is the selective advantage of the favored allele, the frequency, $x(t)$, of the favored allele decreases as the simulation works back in time. For large α , as assumed here, the frequency is approximately

$$x(t) = \frac{1 - \epsilon}{\epsilon + (1 - \epsilon)e^{-\alpha(t - \Delta t)}} \quad (1)$$

(STEPHAN *et al.* 1992). This deterministic x process starts at

TABLE 1
Definitions of symbols

Symbol	Meaning
N	Diploid population size
n	Number of alleles or sequences in the sample
M	Maximum distance (measured in units of recombinational distance) from the neutral locus that a selected locus can be
α	$2Ns$
s	Selective advantage of the favored allele per generation
$x(t)$	Frequency of the selectively favored allele at time t
ϵ	Value of $x(t)$ at the end of the deterministic portion of the selective phase; $1 - \epsilon$ is the value of $x(t)$ at the start of the deterministic portion of the selective phase
Δt	Length of a time step in a selective phase
R	Expected number of crossovers between the neutral locus and selected locus per genome per $2N$ generations
S	Number of segregating sites
$E(T)$	Expected size of a genealogy
Λ_r	Expected number of hitchhiking events per $2N$ generations per recombination unit
η	Duration of the deterministic portion of the selective phase
Λ_{MAX}	The upper bound on Λ_r
$D_{+0.975}$	Upper neutral 95% critical value of Tajima's D
$D_{-0.975}$	Lower neutral 95% critical value of Tajima's D
X_1^2	Goodness-of-fit statistic of the observed number of unique segregating sites and the number of nonunique sites to the expected numbers of these two classes
X_2^2	Goodness-of-fit statistic of the numbers of uniques, doublets, and the remaining frequency classes pooled to the expected numbers of these three classes
X_{all}^2	Goodness-of-fit statistic of the numbers of segregating sites observed and binned to the neutral number of sites expected in these same bins of width <3 sites
$R_{E(T)}$	$E(T)$ relative to its neutral expectation
R_{θ}	Ratio of $\hat{\theta}$ to a typical $\hat{\theta}$ observed in a region of the genome where the rate of crossing over per physical distance is not restricted

frequency $1 - \epsilon$ and ends at ϵ . KAPLAN *et al.* (1989) chose $\epsilon = 5/\alpha$ based on their calculation of the point at which the probability of extinction of the new mutant due to stochastic effects becomes approximately zero. However, we found that under representative sets of parameter values, using $\epsilon = 1/2N$ gave approximately the same $E(T)$, average Tajima's D , and the fraction of $D \leq D_{-0.975}$ (defined below) for single hitchhiking events as did using $\epsilon = 5/\alpha$ and $2N = 10^8$, and simulating the stochastic processes near the boundaries with a Poisson branching process. We used $\epsilon = 1/2N$ to reduce simulation time. Results presented here used $2N = 10^8$ to calculate ϵ . Again, $E(T)$ and Tajima's D , and the fraction of $D \leq D_{-0.975}$ were essentially unchanged when $2N = 10^9$, 10^{10} , or 10^{11} .

During a selective phase, the population of neutral alleles is divided into those linked and those not linked to the favored allele. Within each small increment of time, Δt (chosen to be $1/100\alpha$), one of the neutral alleles linked to the favored allele either can coalesce with another neutral allele linked to the favored allele (Equation 2), or it can recombine onto a chromosome bearing the unfavored allele (Equation 3). If i = the number of neutral alleles linked to the selectively favored allele, the probabilities of these events are

$$\Pr\{\text{coalescent}\} = \frac{\binom{i}{2}}{x(t)} \Delta t, \quad (2)$$

$$\Pr\{\text{recombination}\} = iR(1 - x(t))\Delta t. \quad (3)$$

R is the expected number of recombination events between the selected and neutral loci per $2N$ generations. These equations are based on KAPLAN *et al.*'s (1989) equation 7. Transitions from solid to dashed lines in Figure 2 (right side) represent this type of recombination event in the selective phase.

Similarly, one of the neutral alleles linked to the selectively unfavored allele can coalesce with another neutral allele linked to the unfavored allele, or it can recombine onto a chromosome bearing the favored allele, with probabilities of these events given by Equations 4 and 5, respectively, in which j is the number of neutral alleles linked to the unfavored allele

$$\Pr\{\text{coalescent}\} = \frac{\binom{j}{2}}{1 - x(t)} \Delta t, \quad (4)$$

$$\Pr\{\text{recombination}\} = jRx(t)\Delta t. \quad (5)$$

Transitions from dashed to solid lines in Figure 2 (right side) represent this type of recombination event in the selective phase.

During a selective phase, time (t) is changes in small increments (Δt). After each time step, the probabilities of these four events are calculated according to Equations 2–5 above. The total probability of any of the events occurring in Δt is the sum of these probabilities, and using $\Delta t = 1/100\alpha$ prevented this sum from ever increasing above 0.6. This total probability is subtracted from one to obtain the probability that no event occurs during Δt . The probabilities of zero events are multiplied in each time increment until the product is less than a random number drawn from a uniform distribution between zero and one. At this point, one of the four events is chosen randomly, weighting the choice by the probability of that event at that time.

The selective phase is exited if the genealogy coalesces entirely (at any frequency), or if x decreases below ϵ and there is only one or no neutral allele linked to the selectively favored allele. After exiting a selective phase, the remaining neutral

alleles enter another neutral phase, and the cycle continues until the all of the neutral alleles have coalesced.

After a genealogy is constructed, S (predetermined) neutral mutations are distributed randomly onto the genealogy according to a uniform distribution. The probability a mutation is applied to a particular branch of a genealogy is proportional to the branch's length (in time). These mutations are then projected onto the original sample of neutral alleles depending on the mutations' locations among the branches describing the history of those alleles. Since this algorithm models the infinite sites model, each mutation creates a distinct segregating site at which there are two states. In general, this method follows HUDSON's (1993) third method, such that the statistical analysis in this paper conditions on a predetermined number of segregating sites, not on $4Nu$. We chose this method because S is known for particular data sets, while the true value of $4Nu$ for the loci we examine is unknown.

One assumption of the simulations is that at most one hitchhiking event can occur at any given time (KAPLAN *et al.* 1989). In our model, selected substitutions occur independently according to a time-homogeneous Poisson process. Thus the probability that a second substitution does not occur during a hitchhiking event in progress is the zero term of the Poisson process describing the occurrence of selected substitutions, *i.e.*,

$$p = e^{-2M\Lambda\eta^*} \quad (6)$$

(KAPLAN *et al.* 1989, unnumbered equation p. 894). Λ is the rate at which recurrent selected substitutions occur per $2N$ generations per recombination unit (KAPLAN *et al.* 1989). η^* , the expected time spent in the selective phase (deterministic and stochastic), can be found in Table 2 of KAPLAN *et al.* (1989), except for the $\alpha = 10^7$ case. η^* for this case was determined by repeating their method. One can substitute Λ_{MAX} for Λ , in Equation 6 and solve for Λ_{MAX} , the maximum Λ , for which the chance of a second substitution occurring is less than p . When $p = 0.95$ and $\Lambda_r \leq \Lambda_{MAX}$, there is less than a 5% chance of a second hitchhiking event occurring during the present one. Note that we calculated Λ_{MAX} differently than KAPLAN *et al.* (1989); they calculated it such that the probability of two substitutions occurring at the same time for *all* the selective phases of an entire genealogy was minimized, while we calculate it with respect to each individual selective phase. This is the reason why our Λ_{MAX} values are larger than theirs.

Our simulations of the $n = 2$ case (data not shown) produced results consistent with those of KAPLAN *et al.* (1989), indicating the reliability of our simulations.

The upper (2.5%) and lower (2.5% and 5%) neutral confidence limits of Tajima's D (TAJIMA 1989) and Fu and Li's D and D^* (FU and LI 1993) were determined by examining the distributions of these statistics calculated from 10,000 genealogies simulated without hitchhiking. Fu and Li's D and D^* are test statistics based on deviations from expectations of the numbers of mutations falling on internal and external branches of a neutral genealogy. Fu and Li's D is used when an outgroup is available and their D^* is used when one is not. The critical values of Tajima's D will be represented as $D_{-0.975}$, $D_{+0.025}$, and $D_{-0.95}$. TAJIMA (1989) provided a table of confidence intervals for D based on a beta approximation of the true distribution of D . Although most authors to date have used these approximated critical values to determine whether D was significant in their surveys, we estimated the exact critical values (conditioned on the number of segregating sites) and used them to determine the significance of all observed or simulated values of D discussed here.

TAJIMA's (1989, equation 51) equation for the neutral expectation of the number of sites represented at a particular frequency in a sample of n alleles with S segregating sites was employed to generate three new statistics. In general, these

statistics are the goodness-of-fit of the observed and expected numbers of sites falling into various classes,

$$X^2 = \sum_{i=1}^{\text{no. classes}} \frac{(\text{observed}_i - \text{expected}_i)^2}{\text{expected}_i} \quad (7)$$

The first statistic, X^2_1 , is the goodness-of-fit of the observed number of unique segregating sites and the number of non-unique sites to the expected numbers of these two classes. X^2_2 is the goodness-of-fit statistic of the numbers of uniques, doublets, and the pooled remaining frequency classes to the expected numbers of these three classes. X^2_{all} were constructed to have the expected number of sites between two and three, except for the uniques bin and the highest frequency class bin. The highest frequency class bin was pooled with the second highest frequency class bin. The upper 95% confidence limits of these statistics were determined as described above for Tajima's D , and they were used as one-tailed test statistics.

The 95% confidence interval of the Ewens-Watterson (EWENS 1972; WATTERSON 1978) test statistic F for each sample size and number of alleles in the sample was determined using the algorithm by STEWART (1977). We based our determination of the neutral distribution of F on a computer program by MANLY (1985, pp. 448–450).

RESULTS

In this section, we examine the effect of genetic hitchhiking on the frequency spectrum of segregating sites, quantified by the statistic Tajima's D (TAJIMA 1989). The average D s simulated under the simple hitchhiking model are compared to some example DNA sequence data from natural populations of *D. melanogaster* from genomic regions where the effects of hitchhiking are thought to be detectable (MARTÍN-CAMPOS *et al.* 1992; BEGUN and AQUADRO 1993, 1995; AGUADÉ *et al.* 1994). To make this comparison, a method of calibrating the simulations to the appropriate strength of the hitchhiking effect is developed. Also, the powers of Tajima and others' statistical tests to distinguish the hitchhiking model from neutrality are estimated. Finally, the proportion of times this simulation produced data sets with a $D \leq D_{obs}$ is presented as the probability of observing a particular D_{obs} under the simple hitchhiking model.

The rate of recurrent hitchhiking, Λ_r , and the strength of selection, α , are the two parameters governing the strength of the hitchhiking effect. We investigated the consequences of increasing Λ_r and α on $E(T)$, the expected total size of a genealogy of a sample of alleles measured in units of $2N$ generations. The total size of a genealogy is calculated by summing the lengths of all its branches. Because we are interested in the reduction of $E(T)$, we define relative $E(T)$,

$$R_{E(T)} = \frac{E_{hh}(T)}{E_{neu}(T)}, \quad (8)$$

where $E_{hh}(T)$ is the expected total size of a genealogy under hitchhiking (estimated by the average total size of genealogies from the simulations with hitchhiking) divided by the neutral expectation of the total size of

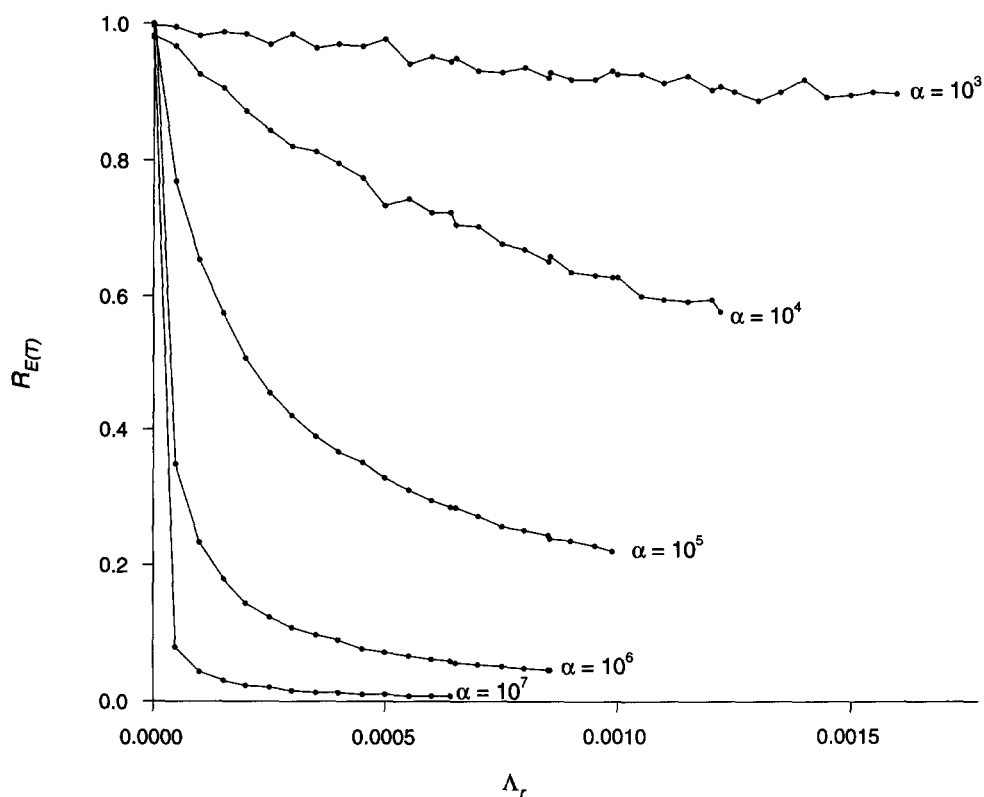


FIGURE 3.— $R_{E(T)}$, the expected total time in a genealogical tree divided by its neutral expectation, plotted against Λ_r . The parameters are $n = 50$, $S = 17$, $\alpha = 10^3, 10^4, 10^5, 10^6$, or 10^7 , and Λ_r ranges from zero to Λ_{MAX} , which varies depending on α . Between 14 and 37 different values of Λ_r were used for each α .

a genealogy [*i.e.*, $E_{neu}(T) = \sum_{i=1}^{n-1} (2/i)$ (WATTERSON 1975)].

To confirm that the simple hitchhiking model does decrease $E(T)$ (and thereby decreases the number of segregating sites), Figure 3 contains a plot of $R_{E(T)}$ against Λ_r . Each series was calculated with a different value of α , *i.e.*, $10^3, 10^4, 10^5, 10^6$, and 10^7 , and between 14 and 37 different values of Λ_r were used to draw these lines. We present the case where $n = 50$ in this figure (and those following) to make it directly comparable to the 50 sequences in the *su(w^o)* data set (AGUADÉ *et al.* 1994). It is evident from Figure 3 that $R_{E(T)}$ decreases as Λ_r increases for a given α . $R_{E(T)}$ also decreases with increasing α for a given Λ_r . This is the same pattern observed by KAPLAN *et al.* (1989) in their Figure 4.

A similar analysis was carried out to determine the effect of increasing Λ_r and α on the average Tajima's D . The results for the $n = 50$ and $S = 17$ case are presented in Figure 4 to reflect the *su(w^o)* data set. There is a pronounced trend toward negative values of average Tajima's D when Λ_r and/or α increase. To demonstrate that this decrease in Tajima's D reflects a skew in the frequency spectrum toward rare variants, we show two examples of frequency spectra in Figure 1. The black bars indicate a neutral frequency spectrum and the gray bars depict a frequency spectrum produced under strong hitchhiking conditions. One thou-

sand genealogies were simulated for this figure with $n = 50$, $S = 17$, $\alpha = 10^6$, and $\Lambda_r = 6.400 \times 10^{-5}$. The vertical axis in Figure 1 indicates the proportion of polymorphic sites at which the rarer of the two mutational states occurred exactly the number of times indicated on the horizontal axis. For example, the proportion of sites occurring only once in a sample (*i.e.*, uniques; bin "1" on the horizontal axis) was ~ 0.25 for neutral simulations and ~ 0.5 for simulations with strong hitchhiking.

Because the values of the parameters α and Λ_r cannot be estimated from DNA survey data, we use $R_{E(T)}$ as an alternative measure of the strength of the hitchhiking effect. Because the expected size of a genealogy is proportional to the expected number of segregating sites, a plausible estimator of $R_{E(T)}$ is the ratio of $\hat{\theta}$ from a region of reduced crossing over relative to $\hat{\theta}$ from a region of normal crossing over. We use $R_{\hat{\theta}}$ (relative $\hat{\theta}$) to denote this observed reduction in $\hat{\theta}$. We took $\hat{\theta}$ in regions of normal crossing over to be 0.006 for North American and European populations because it is a typical value (LANGLEY 1990) and 0.01 for the Zimbabwe population because it is the average $\hat{\theta}$ from the only two loci from regions of normal crossing over surveyed from that population, which appears to have more variation than the others (BEGUN and AQUADRO 1993). Dividing $\hat{\theta}$ from the loci with reduced crossing

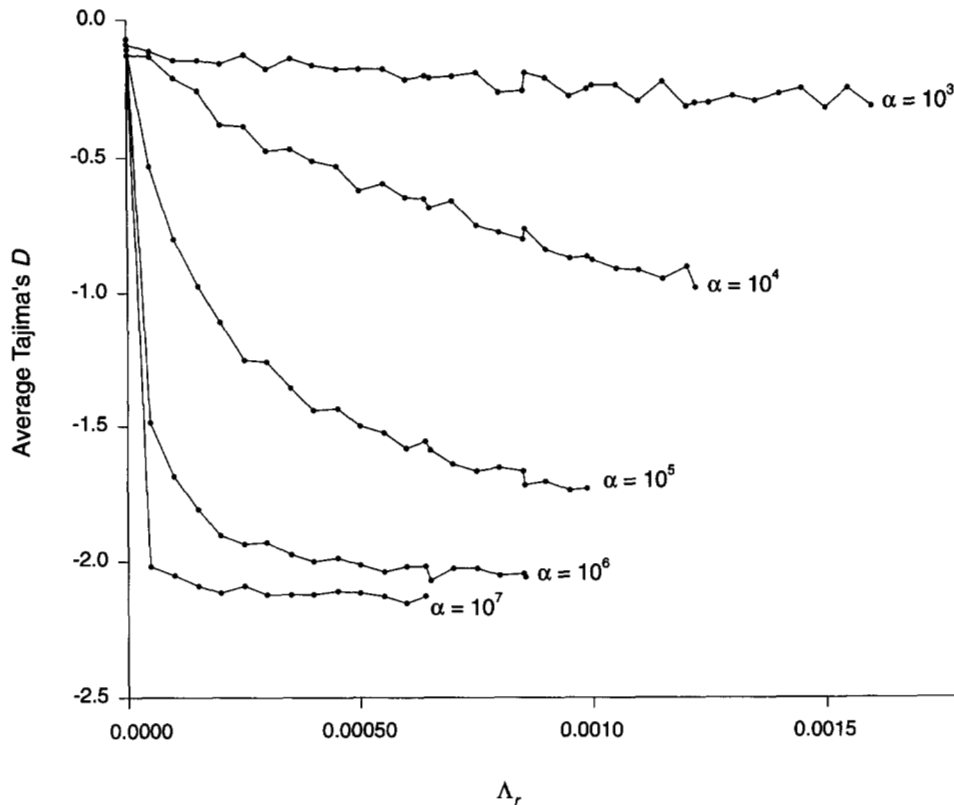


FIGURE 4.—The average value of Tajima's D as a function of Λ_r . The parameters are $n = 50$, $S = 17$, $\alpha = 10^3, 10^4, 10^5, 10^6$, or 10^7 , and Λ_r ranges from zero to Λ_{MAX} , which varies depending on α .

over by these typical $\hat{\theta}$ s gives $R_{\hat{\theta}}$. Table 2 lists $R_{\hat{\theta}}$ for several recent data sets from regions of the *D. melanogaster* genome where crossing over is reduced and thus where the hitchhiking effect is expected to operate (MARTÍN-CAMPOS *et al.* 1992; BEGUN and AQUADRO 1993, 1995; AGUADÉ *et al.* 1994). Included in this table are one example of a locus exhibiting a significant and negative Tajima's D , and the only four published surveys in which Tajima's D could be calculated yet was not significantly different from zero.

Plotted in Figure 5 is average Tajima's D against $R_{E(T)}$. Each data point (larger circles) in this figure is the average D calculated from 1000 genealogies constructed for a different combination of Λ_r , which ranged from zero to Λ_{MAX} , and α , which equaled $10^3, 10^4, 10^5, 10^6$, or 10^7 . These are the same points used to draw Figure 3. Also plotted in this figure are the 95% confidence limits (smaller circles) for each average D . Figure 5 demonstrates that the average Tajima's D decreases as hitchhiking reduces $E(T)$. The ability of hitchhiking to reduce $E(T)$ depends on α and/or Λ_r , per Figure 3. Thus moving left on the horizontal axis is the result of increasing α and/or Λ_r . Because the average D decreases linearly if either α or Λ_r is increased (in this range), $R_{E(T)}$ can be used as a general measure of the strength of the hitchhiking effect. This figure was produced to reflect the *su(w^o)* data set that had $n = 50$ and $S = 17$, but it is typical of all parameters we examined (all permutations of $n = \{10, 30, 50\}$ and $S = \{10, 30, 50\}$). For the *su(w^o)* locus, $R_{\hat{\theta}} = 0.32$ (Table

2). If one takes this as an estimate of $R_{E(T)}$, then according to Figure 5 the expected value of D is -1.52 . This result is listed in Table 2, as is the average Tajima's D s simulated with values of α and Λ_r chosen such that the $R_{E(T)}$ approximately equaled the $R_{\hat{\theta}}$ of four other recent data sets. In spite of these large, negative expected values of D under the hitchhiking model, four loci exhibiting low $R_{\hat{\theta}}$ had D s near zero.

While the above analysis indicates that a large shift in D is expected under hitchhiking strong enough to reduce $\hat{\theta}$ by $R_{\hat{\theta}}$, a more pertinent question concerns the probability of observing a statistically significant negative D given a particular reduction in $R_{\hat{\theta}}$. Figure 6 depicts the outcome of a series of simulations with various hitchhiking parameters. Each circle was obtained with a unique combination of values of α and Λ_r , for a sample size of 50 with 17 segregating sites. The abscissa is the relative reduction in $E(T)$ and the ordinate is the proportion of realizations in which the observed D was less than the $D_{-0.975}$. In general, the trend is toward more cases of $D \leq D_{-0.975}$ as $R_{E(T)}$ decreases (*i.e.*, as the strength of the hitchhiking effect increases).

According to Figure 6, when $R_{E(T)} = 0.32$ (corresponding to the $R_{\hat{\theta}} = 0.32$ exhibited by the *su(w^o)* data) (see Table 2), 0.505 of the genealogies had $D \leq D_{-0.975}$. This result is listed in Table 2 along with the results of similar analyses for four other data sets (MARTÍN-CAMPOS *et al.* 1992; BEGUN and AQUADRO 1993, 1995). Of the four loci in this table with nonsignificant Tajima's D s, *su(s)* had the highest proportion of $D \leq D_{-0.975}$,

TABLE 2
Application of simulation results to relevant data sets

Locus/population	<i>n</i>	<i>S</i>	$\hat{\theta}$	$R_{\hat{\theta}}$	α	Λ_r	Observed Tajima's <i>D</i>	Simulation results				Source
								Proportion of $D \leq$			Proportion of $D \approx D_{\text{obs}}$	
								$D_{-0.975}$	$D_{-0.95}$	$D_{-0.95}$		
<i>y-ac-sc</i> Barcelona	50	9	0.00064	0.11	10^6	2.947×10^{-4}	-2.04	0.652	0.725	0.628	MARTÍN-CAMPOS <i>et al.</i> (1992)	
<i>su(s)</i> N. Carolina	50	10	0.00069	0.12	10^6	2.620×10^{-4}	+0.81	0.676	0.795	0.001	AGUADE <i>et al.</i> (1994)	
<i>su(w^a)</i> N. Carolina	50	17	0.0019	0.32	10^6	6.400×10^{-5}	-0.40	0.505	0.605	0.078	AGUADE <i>et al.</i> (1994)	
<i>y-ac</i> Zimbabwe	50	8	0.0015	0.15	10^6	1.900×10^{-4}	-0.42	0.601	0.701	0.052	BEGUN and AQUADRO (1993; 1995)	
<i>su(f)</i> Zimbabwe	50	6	0.0011	0.11	10^6	2.947×10^{-4}	+0.34	0.605	0.678	0.011	BEGUN and AQUADRO (1993; 1995)	

Included in this table are only those loci and populations for which sufficient variation was found with which to calculate Tajima's *D*. The only statistically significant D_{obs} in this table is *y-ac-sc* from Barcelona. R_{θ} was calculated by dividing θ for each locus by 0.006 for the North Carolina and Barcelona populations or 0.01 for the Zimbabwe populations. The values of α and Λ_r were chosen such that the $R_{\theta(T)}$ simulated with those values equaled R_{θ} . The average simulated Tajima's *D*s, the proportion of simulated *D*s less than the $D_{-0.975}$ and $D_{-0.95}$ critical values, and the proportion of simulated *D*s greater than or equal to the observed value, D_{obs} were based on 1000 iterations.

namely 0.676. The *su(w^a)* locus had the lowest proportion of $D \leq D_{-0.975}$. In spite of these somewhat high probabilities of *D* being less than $D_{-0.975}$ in the simulation, *D* was not observed to be significantly different from zero in four of the five data sets. The proportion of simulated $D \leq D_{-0.95}$ for these loci is also presented in Table 2 to emulate a one-tailed test at the 95% significance level.

If these loci are assumed to be independent, the probability of observing these Tajima test results for these five loci combined can be determined. To do this, we simulated hitchhiking under the parameters relevant to each of the five loci. One Tajima's *D* was picked randomly (without replacement) from each of the five simulated sets of 1000 *D*s. The times one or none of these picked *D*s was less than or equal to the relevant critical value was counted. The probability that a configuration of one significant and four nonsignificant or of five nonsignificant Tajima's *D*s appeared in the hitchhiking simulations was 0.084 for $D_{-0.975}$ and 0.024 for $D_{-0.95}$.

Another way to investigate the utility of such statistics as Tajima's *D* in detecting the impact of hitchhiking is to calculate the probability of observing a *D* greater than or equal to the value observed in a particular data set from a natural population, D_{obs} . This probability was determined in a hitchhiking simulation with a sample size and number of segregating sites equal to those in a specific data set, and with sets of values of α and Λ_r that yield a reduction in $E(T)$ comparable to that estimated by R_{θ} . The results are presented in the second-to-last column of Table 2. Except for *y-ac-sc* from Barcelona, in which a significant value of *D* was observed, the proportion of times the simulation produced genealogies with $D \geq D_{\text{obs}}$ was always less than 0.078. The number of times *D* simulated under the *su(w^a)* parameters was greater than or equal to D_{obs} for this data set was also plotted in Figure 6 (squares) over a range of $R_{\theta(T)}$. One can perform an analysis of the full table, to estimate the probability of five loci, under the hitchhiking model calibrated as in Table 2, having *D*s equal to or greater than those observed in the real data sets. This analysis assumes that the loci are independent. The number of times each of five randomly chosen *D*s (one from each of the 1000 *D*s simulated under the parameters appropriate for each of the loci) was greater than or equal to the D_{obs} was zero.

We also investigated the proportion of times several other statistics fell outside their neutral critical values and compared the results with the above results for Tajima's *D*. The outcome of the analysis for X_1^2 , X_2^2 , and X_{all}^2 is plotted against $R_{\theta(T)}$ in Figure 7. Tajima's *D* and X_1^2 have the highest and roughly the same proportion of occurrences beyond their neutral critical values. X_2^2 has the next highest, and X_{all}^2 generally has the lowest proportion. The proportion of times Fu and Li's D^* and the Ewens-Watterson *F* occurred beyond their

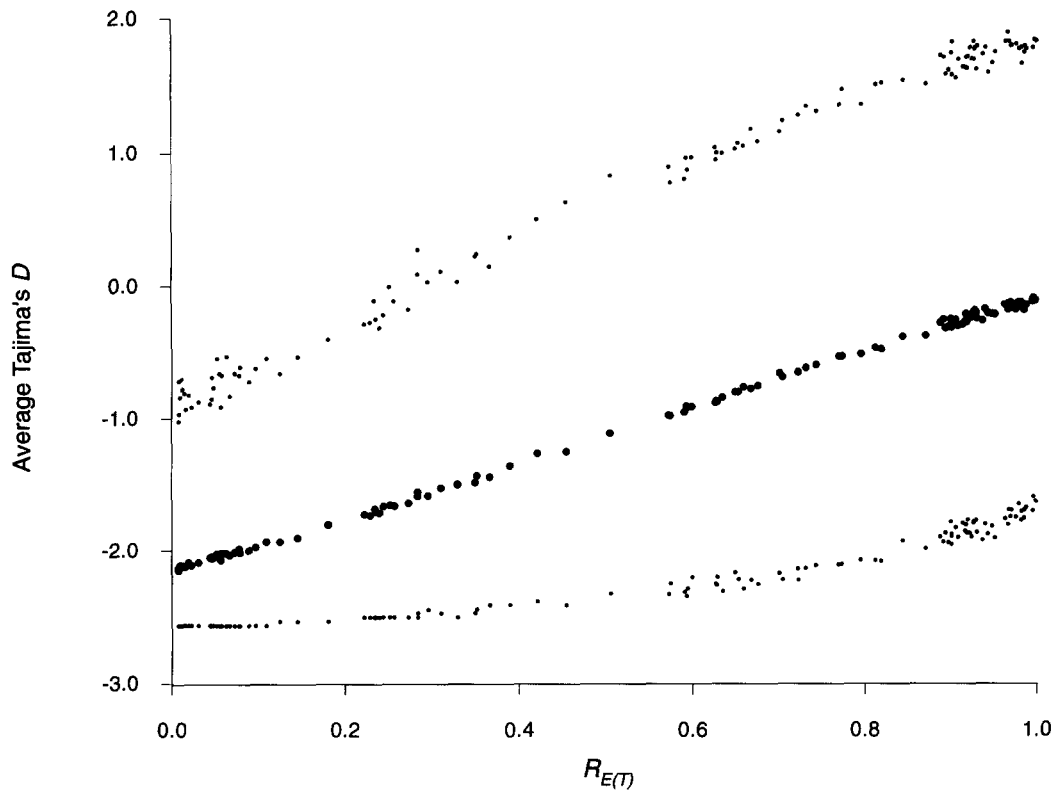


FIGURE 5.—The average value of Tajima's D (●) as a function of $R_{E(T)}$, the expected total time in a genealogical tree divided by its neutral expectation. The parameters are $n = 50$, $S = 17$, $\alpha = 10^3, 10^4, 10^5, 10^6$, or 10^7 , and Λ , ranges from zero to Λ_{MAX} , which varies depending on α . The small circles indicate the upper and lower 95% confidence limits of each average Tajima's D .

neutral critical values are plotted in Figure 8. The same information for Tajima's D is also plotted there for comparison. The results are essentially the same for these three statistics. Because they both are based on the number of unique sites expected under neutrality, X_1^2 (Figure 7) and Fu and Li's D^* (Figure 8) fall beyond their neutral critical values a similar number of times. The same analysis was applied to Fu and Li's D (to be used when an outgroup is available) (data not shown), and the proportion of times it falls beneath the negative critical value is slightly less than that for D^* .

DISCUSSION

Before this investigation, the effects of genetic hitchhiking on DNA sequence polymorphism could be analyzed only in terms of the predictions of MAYNARD SMITH and HAIGH (1974), STEPHAN *et al.* (1992), and KAPLAN *et al.* (1989). The first two described the hitchhiking effect on heterozygosity, and the third described the hitchhiking effect on the expected number of segregating sites in a sample. An extreme reduction in both these quantities has often been observed in regions of the *D. melanogaster* genome where crossing over per physical length was reduced, consistent with the predictions of these analyses (AGUADÉ *et al.* 1989, 1994; MIYASHITA 1990; BEGUN and AQUADRO 1991, 1993, 1995; BERRY *et al.* 1991; MARTÍN-CAMPOS *et al.* 1992; LANGLEY *et*

al. 1993). Most observers have viewed these consistently repeated empirical results as strong support for the hitchhiking effect model, especially in those cases where no corresponding reduction in interspecific divergence was observed.

However, there were several limitations of the analyses of MAYNARD SMITH and HAIGH (1974), STEPHAN *et al.* (1992), and KAPLAN *et al.* (1989). First, these could not provide the stochastic distributional properties of their statistics; they only gave expectations. Second, with regard to the sample properties of their predictions, MAYNARD SMITH and HAIGH (1974) and STEPHAN *et al.* (1992) only provided results for the whole population, and KAPLAN *et al.* (1989) only analyzed the sample size $n = 2$ case. Third, none of these authors was able to investigate the properties of higher order statistics such as Tajima's D that also might be subject to the hitchhiking effect.

Consequently, the empirical studies reporting such higher order statistics as Tajima's D from their observed data were unable to determine with any great degree of certainty whether their results supported the hitchhiking effect model. According to verbal arguments, hitchhiking results in a skew toward rare variants in the frequency spectrum (AGUADÉ *et al.* 1989; HUDSON 1990). Thus when Tajima's D was observed to be significantly negative, this was attributed to the hitchhiking effect. Data sets in which sufficient variation was

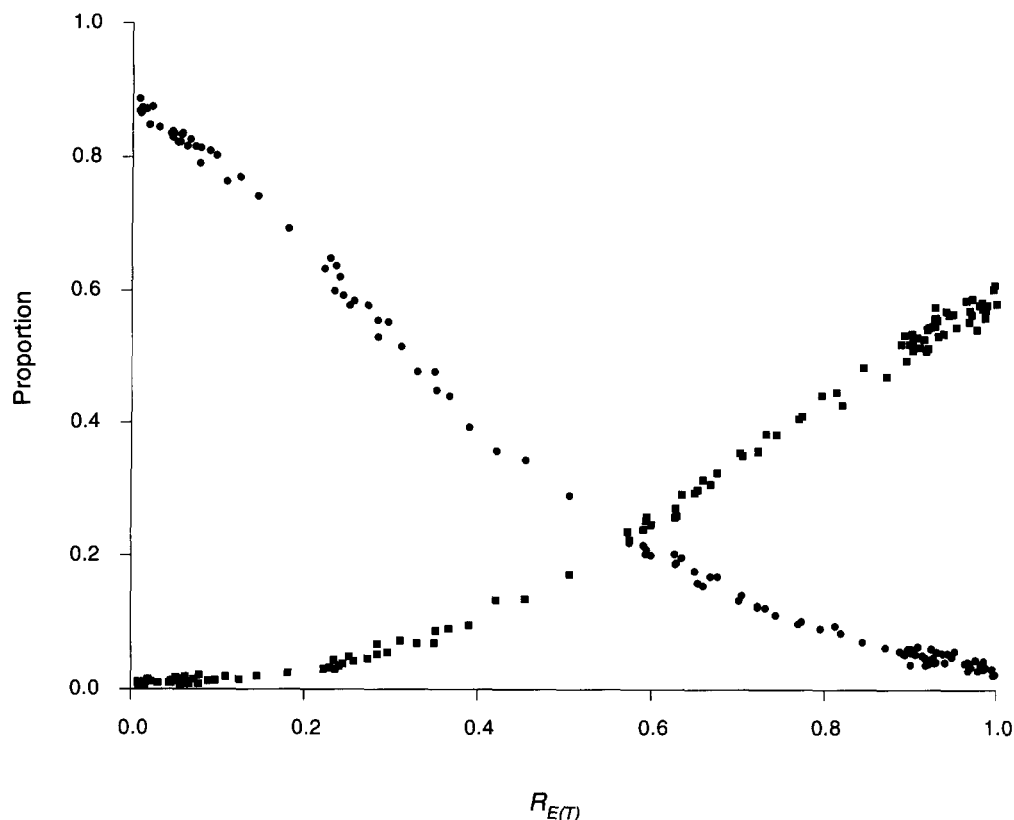


FIGURE 6.—The proportion of simulations in which Tajima's D is $\leq D_{-0.975}$ (●) plotted against $R_{E(T)}$ and the proportion of simulations in which Tajima's D equaled or fell above -0.40 , the D observed in the $su(w^e)$ data set (■). The parameters are $n = 50$, $S = 17$, $\alpha = 10^3, 10^4, 10^5, 10^6$, or 10^7 , and Λ_r ranges from zero to Λ_{MAX} , which varies depending on α .

available to determine Tajima's D sometimes had a significantly negative D , such as some of the populations surveyed by MARTÍN-CAMPOS *et al.* (1992) and AGUADÉ *et al.* (1989).

In contrast, several other studies found the predicted reductions in nucleotide diversity and in the number of segregating sites, yet they did not find any significant deviation of Tajima's D from zero (MARTÍN-CAMPOS *et al.* 1992; BEGUN and AQUADRO 1993, 1995; AGUADÉ *et al.* 1994). This raised several related questions: (1) Is the expected value of Tajima's D actually negative under the simple hitchhiking model? (2) Is Tajima's D statistic powerful enough to detect the hitchhiking effect on the frequency spectrum? (3) Under the simple hitchhiking model, how likely is the observed Tajima's D value in a particular data set from a natural population? The present investigation answered each of these questions.

To answer these questions, we first had to develop a measure of the strength of the hitchhiking effect that would be relevant to our simulations and measurable from data from natural populations. Although Λ_r and α determine the strength of the hitchhiking effect in the simulation, they cannot be estimated from DNA sequence data. Thus we used $R_{E(T)}$, and estimated it with $R_{\hat{\theta}}$, which can be calculated from empirical data. That the strength of the hitchhiking effect can be esti-

mated by $R_{\hat{\theta}}$ over a wide range of values of Λ_r and α is a fortunate and potentially useful finding of this paper. However, this estimator is of limited accuracy because WATTERSON's (1975) $\hat{\theta}$ estimator has a large variance.

Similar to the use of $R_{E(T)}$ as a proxy for knowledge of Λ_r and α is the finding of WIEHE and STEPHAN (1993, equation 5) that $E(T)$ may be approximated by a simple function of the product of Λ_r and α for the sample size $n = 2$. Our simulation results support their analytic approximation and showed that an analogous relationship exists for larger sample sizes (data not shown).

With regard to question (1) above, Figures 3 and 4 show that Tajima's D is indeed expected to be much less than zero when the strength of the hitchhiking effect is sufficient to decrease $R_{E(T)}$ to the observed $R_{\hat{\theta}}$ listed in Table 2. According to these figures and Table 2, when $n = 50$, $S = 17$, *i.e.*, the parameters of the $su(w^e)$ data set of AGUADÉ *et al.* (1994), Tajima's D is expected to be -1.52 . The observed value was only -0.40 . Similar contrasts between D_{obs} and the average simulated D for other relevant loci are presented in Table 2.

To answer the second question, we investigated the power of Tajima's test. The method we used to investigate the power was to calculate the number of times Tajima's D fell beneath the lower 95% neutral critical value of D , $D_{-0.975}$. The fraction of such cases could be

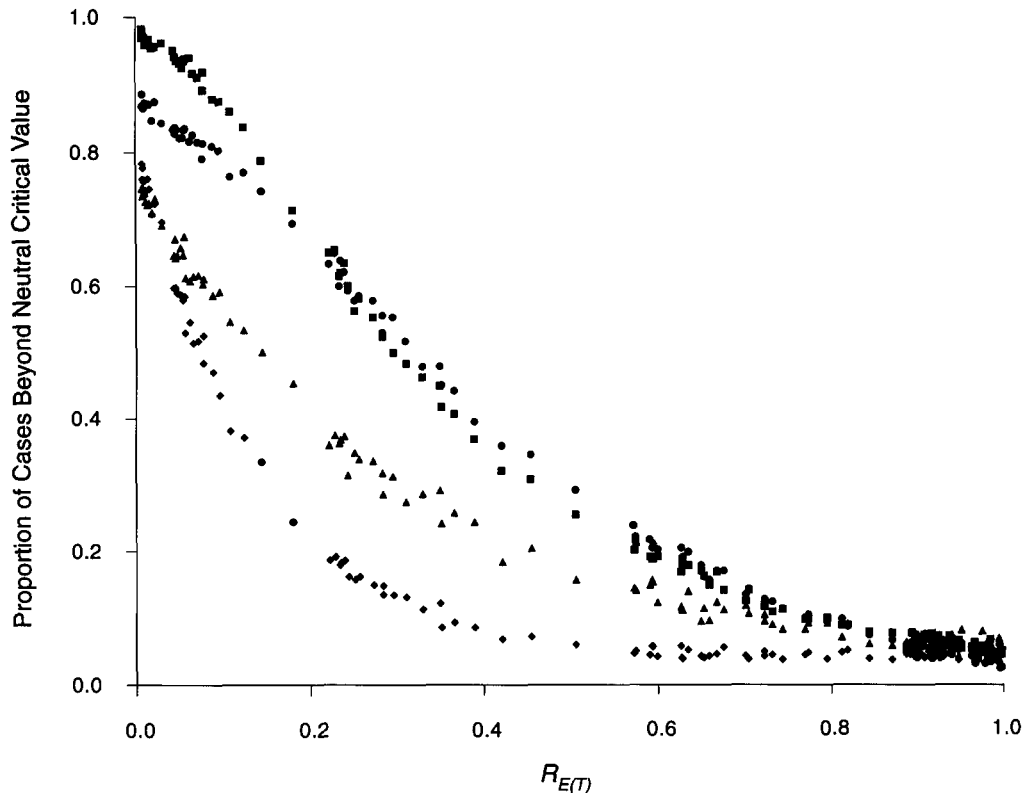


FIGURE 7.—The proportion of cases in which Tajima's D (●), X_1^2 (■), X_2^2 (▲), and X_{all}^2 (◆) fell beyond their 95% neutral critical values plotted against $R_{E(T)}$. The parameters are $n = 50$, $S = 17$, $\alpha = 10^3, 10^4, 10^5, 10^6$, or 10^7 , and Λ , ranges from zero to Λ_{MAX} , which varies depending on α .

viewed as the power of Tajima's test with one qualification. Strictly speaking, power is the probability of rejecting the null hypothesis (neutrality in this instance) given that it is false, irrespective of the direction of the rejection, *i.e.*, it could be above the upper critical value or below the lower critical value. We limited the results presented here to the number of cases beneath the lower critical value. This was justified by two arguments. First, in very few strong hitchhiking simulations did the D s values fall above the upper critical value when $R_{E(T)}$ was around values relevant to this paper. Second, our $D \leq D_{-0.975}$ results can be interpreted as the "power of Tajima's test to detect the hitchhiking effect" at the 2.5% significance level for a one-tailed Tajima test, since the hitchhiking effect on Tajima's D is evidently directional. This makes our two-tailed test at the 5% significance level conservative. According to a one-tailed test at the 5% significance level, the $D \leq D_{-0.95}$ range from ~60% to nearly 80% (Table 2).

According to Figure 6, a somewhat high fraction (over 50%) of the simulations yielded $D \leq D_{-0.975}$ when $n = 50$, $S = 17$, and $R_{E(T)} = 0.32$. These are the parameters from the $su(w^a)$ data (see Table 2). Interpreting this fraction as the power of Tajima's test to detect the simple hitchhiking model, for the parameter values appropriate for the relevant data, the observation of a nonsignificant value of Tajima's D indicates that the

hitchhiking effect model as it is currently formulated is unlikely to explain the data. A similar argument applies to three other data sets reported in Table 2. This power analysis is not limited to the parameter space considered in Table 2. A particular $R_{E(T)}$ can be produced from many different combinations of α and Λ , other than those chosen for Table 2 as examples. Furthermore, Figure 5 shows the power of Tajima's test for $n = 50$, $S = 17$, and a complete range of $R_{E(T)}$. This is relevant because the accuracy of $R_{E(T)}$ is unknown, and because $R_{E(T)}$ will change if calculated by comparing an observed $\hat{\theta}$ to something other than the $\hat{\theta}$ from a region of normal crossing over which we used. If $\hat{\theta}$ values from these regions are shown to be reduced already due to some form of selection, then our power results are conservative.

Third, the probability of observing a particular value of Tajima's D was used as another method of evaluating the simple hitchhiking model. The results are presented in Table 2 for five recent data sets. For the four studies with nonsignificant D values in Table 2, the fraction of $D \geq D_{obs}$ is small (all are ≤ 0.078), so that were the model correct observing these D values would be very unlikely. The fraction of $D \geq D_{obs}$ for the $su(w^a)$ parameters is also plotted in Figure 6 for a range of $R_{E(T)}$ to demonstrate that these values decrease if a less conservative estimate of $R_{E(T)}$ is used. We base our con-

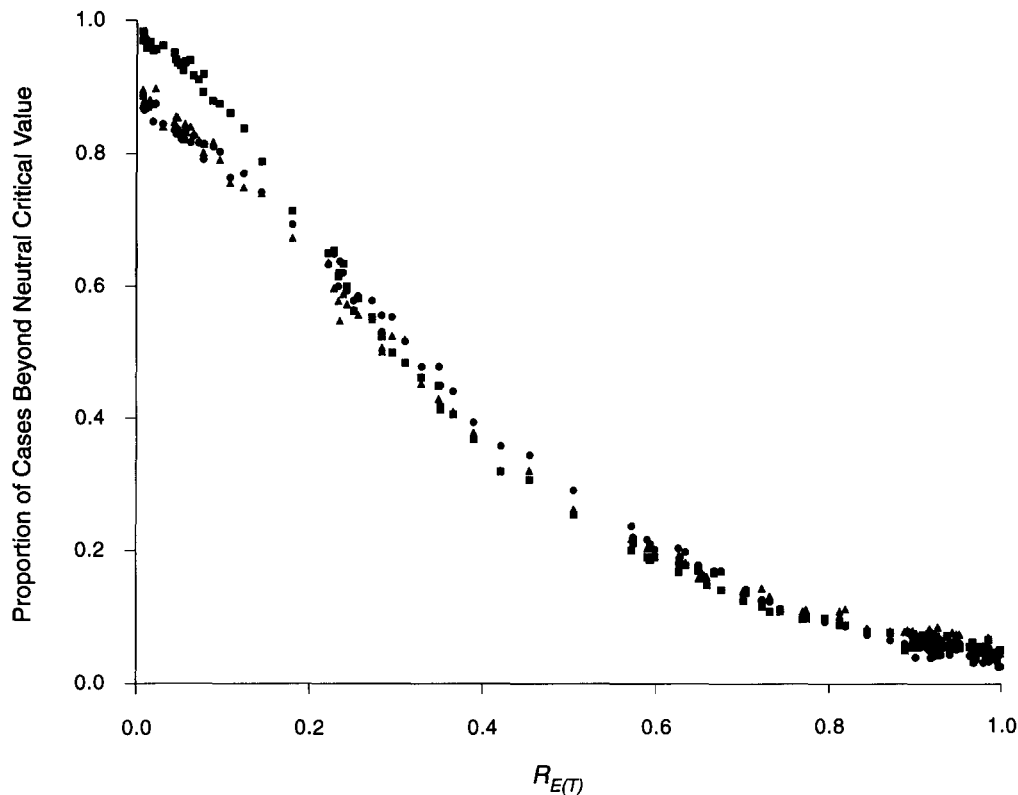


FIGURE 8.—The proportion of cases in which Tajima's D (●), Fu and Li's (1993) D^* (■), and the Ewens-Watterson F statistic (▲) fell beyond their 95% critical values plotted against $R_{E(T)}$. The parameters are $n = 50$, $S = 17$, $\alpha = 10^3, 10^4, 10^5, 10^6$, or 10^7 , and Λ , ranges from zero to Λ_{MAX} , which varies depending on α .

clusion against the simple hitchhiking model primarily on this analysis.

In addition to these analyses of the loci considered separately, we calculated the probabilities of observing all the data presented in Table 2 under the hitchhiking model. As mentioned above, we calculated the probability of observing four or five loci with nonsignificant Tajima's D s under the relevant hitchhiking parameters. These probabilities were 0.084 and 0.024 for $D_{-0.975}$ and $D_{-0.95}$, respectively. No cases of five loci with simulated D s being greater than or equal to D_{obs} were produced in this analysis. This result will stand even if more loci with significant D_{obs} are discovered, according to an analysis in which the case of a significant D_{obs} is represented four times as opposed to just once. In other words, the probability of observing these data sets under the hitchhiking model is extremely small.

We also investigated the powers of several other tests for a skewed frequency spectrum. Tajima's test was generally at least as powerful as Fu and Li's tests based on their D and D^* , the Ewens-Watterson test using F , and three tests using statistics proposed here that describe the goodness-of-fit of observed to predicted portions of the frequency spectrum of segregating sites. While these other statistics may have superior properties in testing other alternatives to the neutral theory than this version of hitchhiking, Tajima's D appears to be a good choice when investigating the simple hitchhiking

model. It should be noted that the critical values in TAJIMA (1989) are crude approximations that can differ from our simulated estimates of the critical values by over 10% in some cases. For example, Tajima reported $D_{-0.975} = -1.800$ for $n = 50$ and all values of S , while we estimated $D_{-0.975} = -1.623$ for $n = 50$, $S = 50$, and $D_{-0.975} = -1.639$ for $n = 50$, $S = 17$.

The fundamental conclusion of this paper is that the simple hitchhiking model of MAYNARD SMITH and HAIGH (1974), KAPLAN *et al.* (1989), and STEPHAN *et al.* (1992) must be reconsidered as the sole explanation of the observed reduction in the per site polymorphism in regions of the *D. melanogaster* genome where the rate of crossing over per physical length is reduced. This conclusion is based on the analysis of surveys where the level of polymorphism is sufficient to apply statistics such as Tajima's D (see Table 2), but possibly applies to other surveys (*e.g.*, BERRY *et al.* 1991; LANGLEY *et al.* 1993) that lack sufficient polymorphism to calculate D and thus were not included. This conclusion may not apply to *y-ac-sc* that did exhibit a significant Tajima's D in several populations (AGUADÉ *et al.* 1989; MARTÍN-CAMPOS *et al.* 1992).

Two important assumptions of this model are worthy of reexamination by future hitchhiking models. First, future models could relax the assumption of the simple hitchhiking model that selected substitutions are separate, independent events. The independence of hitch-

hiking events was assumed for technical reasons and has no biological basis (KAPLAN *et al.* 1989). Second, the assumption that hitchhiking events occur uniformly throughout a species' geographical range should also be reconsidered, because STEPHAN and MITCHELL (1992) found fixed differences between different populations of *D. ananassae*, and BEGUN and AQUADRO (1993) found nearly fixed differences between populations of *D. melanogaster*. Both were in genomic regions where the hitchhiking effect was predicted, suggesting that separate hitchhiking events had occurred in different portions of the geographic ranges of these species. Whether these or any modifications to the simple hitchhiking model can explain the observed reduction in nucleotide diversity and in the number of segregating sites without a reduction in Tajima's *D* is a matter of speculation.

Because the simple directional selection or substitution hitchhiking model has been found to be inconsistent with the data, its successors might use alternative types of selection. There are only two published examples to date. First, GILLESPIE (1994) analyzed models of fluctuating selection coefficients and found this could produce reduced heterozygosity without a decrease in Tajima's *D*. Second, CHARLESWORTH *et al.* (1993; CHARLESWORTH 1994) proposed a model of hitchhiking by negative or background selection. Standing variation is reduced in populations by selective removal of linked deleterious alleles. This model predicted a slightly skewed frequency spectrum, yet these authors do not believe this skew would be detectable in a sample of alleles from a population, according to unpublished results (B. CHARLESWORTH, personal communication; see also HUDSON 1994). However, CHARLESWORTH *et al.* (1993) were skeptical that biologically reasonable parameter values for their model could explain the extreme reduction of R_0 observed at the loci such as *y-ac-sc*.

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